

Load disturbance observer-based control method for sensorless PMSM drive

ISSN 1751-8660

Received on 17th November 2015

Revised on 3rd April 2016

Accepted on 21st April 2016

doi: 10.1049/iet-epa.2015.0550

www.ietdl.org

Lu Xiaoquan ✉, Lin Heyun, Han Junlin

Engineering Research Center for Motion Control of MOE, Southeast University, Nanjing 210096, Jiangsu Province, People's Republic of China

✉ E-mail: seueelab_lxq@163.com

Abstract: This study proposes a sensorless control method for the permanent magnet synchronous machine (PMSM) drive that is robust against load torque variations. The proposed method is based on the disturbance observer-based control (DOBC) method, and involves the use of a back electromotive force observer and a torque observer to estimate rotor position and compensate for load torque disturbance, respectively. This mechanism is simple to implement and requires no state vector derivation. The gains of the two observers are carefully selected to improve estimation accuracy and dynamic performance. The performance of the proposed load DOBC for sensorless PMSM drives was tested through a simulation, where the sensorless estimation error was considered to determine the adjustment laws for the proposed observer gains. Furthermore, results of evaluative experiments verified the effectiveness of the proposed method for high-performance sensorless control and torque disturbance rejection for the PMSM drive.

1 Introduction

The permanent magnet synchronous machine (PMSM) has attracted considerable research interest due to its brushless structure, high efficiency and large torque density [1]. However, the presence of a mechanical speed sensor is a major disadvantage of the PMSM control system: it affects systematic reliability, increases manufacturing cost and restricts the environments in which the machine can be implemented. Sensorless PMSM drives offer a promising solution to these problems by estimating the speed and position of the rotor by using measurable signals, such as current and terminal voltage.

Prevalent sensorless PMSM control algorithms can be categorised into saliency-based estimation and model-based observation. Saliency-based estimation algorithms involve obtaining rotor position by using position-related inductance response under high-frequency voltage or current injection, which are suitable for standstill and startup regions [2–5]. Model-based observers provide satisfactory sensorless control methods for middle- and high-speed regions. The extended Kalman filter determines rotor speed and position by estimating the recursive optimal state of the PMSM [6]. This algorithm is computationally intensive, and causes problems in the selection of covariance matrices and initial values. Non-linear speed observers [7–11] involve the use of stator currents to estimate rotor speed based on mathematical models of machines, and present a promising solution to sensorless control. However, modelling these observers requires the derivation of the state current, which renders estimation results more sensitive to current noise. Another widely used sensorless control strategy for the PMSM system is the sliding mode observer (SMO), which is based on back electromotive force (EMF) or flux linkage estimations [12–15]. The SMO delivers high accuracy, robust parameters and excellent dynamic resistance. However, it requires state current derivation, and inevitably runs into the chattering problem due to the saturation function.

Disturbance observer-based control (DOBC) methods [16–21] have been applied to some systems, such as the servo system and robot manipulation, for better robustness against system disturbance, which is deduced by a deliberately designed disturbance observer and compensated for by appropriate feedback. The design of DOBC is very easy to master by

engineers, and the construction of the disturbance observer helps avoid state vector derivation and the consequent noise found in traditional observers. DOBC methods are divided into linear and non-linear sorts according to the characteristic of the object plant. In some cases, the disturbance observer design of a non-linear system can be applied to a linear prototype.

Sensorless control based on the disturbance observer has been studied in [22–25]. Furthermore, a considerable amount of research has been devoted to improve load torque robustness for PMSM systems with mechanical sensors [26–28] as well as their sensorless counterparts [29]. However, the effect of the speed estimation error on torque disturbance rejection and systematic performance in sensorless methods has largely been ignored.

This paper proposes load DOBC (LDOBC) for sensorless PMSM drives. In order to obtain rotor speed and position, the back EMF, considered as generalised disturbance, is estimated according to the disturbance observer theory. Another disturbance observer is employed to estimate the load torque, which is compensated for in the output of the speed loop proportional integral (PI) controller. Global asymptotic stability is guaranteed by Lyapunov stability analysis, and the influence of the speed estimation error on the load torque observer is taken into account. The adjustment laws for the back EMF and torque observer gains are determined through simulations to obtain the most satisfactory overall performance. Experiments were conducted to verify the effectiveness of the proposed method.

2 Disturbance observer-based control

2.1 Underlying concept

A linear plant in the time domain can be described as follows

$$\begin{cases} \dot{x} = Ax + B_0 u + B_1 d \\ y = Cx \end{cases} \quad (1)$$

where x , u and d are the state vector, input and external disturbance, respectively, and A , B_0 , B_1 and C represent the coefficient matrices of the state functions.

The general design procedure of DOBC for a linear plant is as follows: A baseline controller is first designed to achieve the required performance under the assumption that the disturbance is measurable. A disturbance observer is then designed for disturbance estimation and integrated into a baseline controller to compensate for the disturbance in the control law in the opposite direction. The block diagram of DOBC is shown in Fig. 1. Suppose the control law of the baseline controller is given by the following equation

$$u^* = f(x) \quad (2)$$

It is assumed the disturbance affects the state input through an identical channel when $B_0 = B_1$. The composite control law of DOBC is given by the following equation

$$u = u^* + \hat{d} = f(x) + \hat{d} \quad (3)$$

Since the design procedures for the controller and the disturbance observer are separate, DOBC is highly flexible with regard to combining with any control scheme to improve its ability to resist disturbance.

2.2 Linear disturbance observer

When a disturbance is slowly time varying, it satisfies

$$\dot{d} = 0, \quad (4)$$

the estimation error is derived from

$$e_d = \hat{d} - d \quad (5)$$

where the superscript $\hat{\cdot}$ denotes the estimated value. From (1)

$$B_1 \dot{d} = \dot{x} - Ax - B_0 u \quad (6)$$

The initial disturbance observer is thus constructed as

$$\dot{\hat{d}} = -L(B_1 \hat{d} - B_1 d) = -LB_1 \hat{d} + L(\dot{x} - Ax - B_0 u) \quad (7)$$

However, the derivative of the state vector cannot be directly measured, which is inconvenient in practical applications. Therefore, the disturbance observer is deformed into

$$\dot{\hat{d}} - L\dot{x} = -LB_1 \hat{d} - L(Ax + B_0 u) \quad (8)$$

By defining an intermediate variable $z = \hat{d} - Lx$, the refined

disturbance observer [16] is formulated as

$$\begin{cases} \dot{z} = -LB_1(z + Lx) - L(Ax + B_0 u) \\ \hat{d} = z + Lx \end{cases} \quad (9)$$

The refined disturbance observer is better suited for practical implementation because it circumvents the need for the derivation of the state vector, and its response speed to disturbance can be easily adjusted by altering the observer gain.

3 PMSM model

The current state and the output equations of a surface-mounted PMSM – a typical non-linear system – can be simplified into the following linear expression

$$\begin{cases} \dot{x}_e = A_e x_e + B_{e0} u_e + B_{e1} d_e \\ y_e = C_e x_e \end{cases} \quad (10)$$

where

$$\begin{aligned} A_e &= \begin{pmatrix} -\frac{R_s}{L_s} & 0 \\ 0 & -\frac{R_s}{L_s} \end{pmatrix}, \quad B_{e0} = \begin{pmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \end{pmatrix} \\ B_{e1} &= \begin{pmatrix} -\frac{1}{L_s} & 0 \\ 0 & -\frac{1}{L_s} \end{pmatrix}, \quad C_e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ x_e &= \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix}, \quad u_e = \begin{pmatrix} u_\alpha \\ u_\beta \end{pmatrix}, \quad d_e = \begin{pmatrix} e_\alpha \\ e_\beta \end{pmatrix}, \quad y_e = \begin{pmatrix} i_\alpha \\ i_\beta \end{pmatrix} \end{aligned}$$

L_s and R_s in the coefficient matrices represent stator inductance and resistance, respectively. Voltages $(u_\alpha \ u_\beta)^T$ and back EMF $(e_\alpha \ e_\beta)^T$ are selected as the state inputs u_e and the disturbance d_e , respectively, and the $\alpha\beta$ -axis currents $(i_\alpha \ i_\beta)^T$ are selected as both the state vectors x_e and the outputs y_e .

Moreover, the state and the output equations based on the mechanical motion equation of the surface-mounted PMSM are linear, and can hence be presented as

$$\begin{cases} \dot{\omega}_r = -\frac{B}{J} \omega_r + \frac{3}{2} \frac{p\psi_f}{J} i_q - \frac{T_L}{J} \\ y = \omega_r \end{cases} \quad (11)$$

where B is the viscous friction coefficient, J is the moment of inertia, p is the number of pole pairs, ψ_f is the PM flux linkage and T_L is the load torque. In this model, mechanical rotor speed ω_r and q -axis stator current i_q are selected as the state vector and input, respectively.

4 Disturbance observer-based position estimation

4.1 Back EMF observer

To implement sensorless control in the PMSM, back EMF is used as an extended disturbance to be estimated by the refined disturbance observer in (9). The derivative of the back EMF during a sampling period is assumed to be zero. Considered as generalised disturbance, the back EMF satisfies the slowly time varying demand. From (9) and (10), the back EMF observer based on the

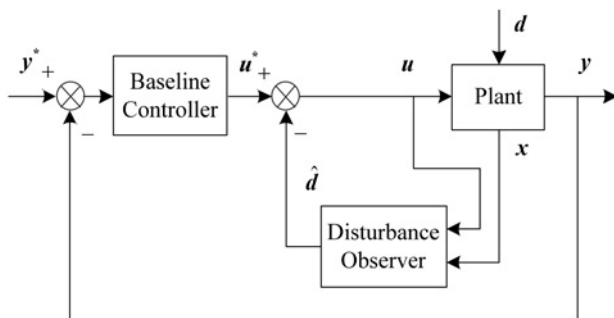


Fig. 1 Structure of DOBC

calculated by the following equation

$$\theta_{\text{comp}} = \arctan\left(-\frac{L_s p \omega_{\text{rated}}}{l_{\text{base}}}\right) \quad (26)$$

Equation (26) shows that a reduction in the value of I_{e_base} can improve the accuracy of the estimation of rotor position by diminishing the compensation angle. As long as the base value of the back EMF observer gain is determined, the compensation angle is a constant value regardless of speed variation. However, when the PMSM is at low speed, the estimation result is not as accurate as expected due to inverter non-linearities and stator resistance variation. In this situation, voltage/frequency (V/F) control is adopted.

At the same time, the disturbance observer is equivalent to a low-pass filter for high-frequency noise, the cutoff frequency of which depends on observer gain L according to (21). By properly adjusting observer gain, undesirable measurement error can be eliminated at the cost of a dynamic response delay in estimated disturbance. For the proposed back EMF observer, an increase in the value of l_{e_base} reduces fluctuation in estimated speed but increases the compensation angle. Therefore, the base value of the back EMF observer gain l_{e_base} should be carefully selected to guarantee values of both compensation angle and estimated speed that are conducive to the satisfactory performance of the sensorless PMSM drive.

5 LDOBC

5.1 Load torque observer considering speed estimation error

The PI controller has a simple structure and is easy to implement for PMSM speed control. However, the disturbance such as a varying load torque degrades the motion accuracy of a gain-fixed PI controller. As a mechanical value, load torque T_L is generally assumed to be slowly time varying in a sample period. Moreover, the slowly time varying request of the disturbance observer can be relaxed. For a sudden load step condition, the initial singular point does not affect the stability of subsequent operation. Therefore, the load torque can be estimated using the disturbance observer scheme, and integrated into the PI controller to improve its dynamic response. The load torque observer is established based on the mechanical motion (11), where mechanical rotor speed ω_r is estimated through the proposed back EMF observer detailed in the previous section. The load torque estimation accuracy is inevitably affected by speed estimation error $\Delta\omega_r$.

$$\Delta\omega_r = \hat{\omega}_r - \omega_r \quad (27)$$

Substituting (27) into (11), the mechanical motion equation of the surface-mounted PMSM under sensorless control is rewritten as

$$\begin{aligned}\dot{\omega}_r = & -\frac{B}{J}\dot{\omega}_r + \frac{3}{2}\frac{p\psi_f}{J}i_q \\ & -\frac{1}{J}(T_L - B\Delta\omega_r - J\Delta\dot{\omega}_r)\end{aligned}\quad (28)$$

The disturbance estimated by the load torque observer established in (28) is a combination of the real load torque and the differences due to the speed estimation error. The speed estimation error can be categorised into a direct current (DC) component $\Delta\omega_{r_DC}$ and a high frequency (HF) component $\Delta\omega_{r_HF}$

$$\Delta\omega_r = \Delta\omega_{r\text{ DC}} + \Delta\omega_{r\text{ HF}} \quad (29)$$

The influence of the HF component can be restrained by adjusting the bandwidth of the torque observer. Therefore, the DC component, which satisfies $\Delta\dot{\omega}_{\text{DC}} = 0$, is the main cause of load

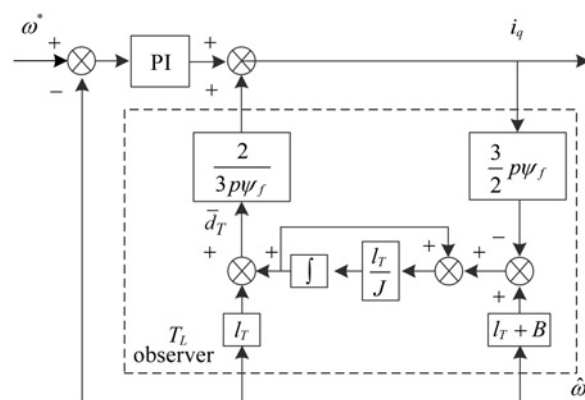


Fig. 3 Block diagram of the proposed LDOBC

torque estimation error

$$\bar{d}_T = T_L - B\Delta\omega_{r,DC} \quad (30)$$

The mechanical motion (28) can be expressed as

$$\dot{\mathbf{x}}_{\text{T}} = \mathbf{A}_{\text{T}}\mathbf{x}_{\text{T}} + \mathbf{B}_{\text{T}_0}\mathbf{u}_{\text{T}} + \mathbf{B}_{\text{T}}\bar{\mathbf{d}}_{\text{T}} \quad (31)$$

where

$$\begin{cases} A_T = -\frac{B}{J}, & x_T = \hat{\omega}_r \\ B_{T0} = \frac{3}{2} \frac{p\psi_f}{J}, & u_T = i_q \\ B_{T1} = -\frac{1}{J}, & \bar{d}_T = T_L - B\Delta\omega_{r,DC} \end{cases}$$

From (9) and (31), the load torque observer considering the speed estimation error is given by the following equation

$$\begin{cases} \dot{\bar{z}}_{\text{T}} = \frac{l_{\text{T}}}{J} \left[\bar{z}_{\text{T}} + (l_{\text{T}} + B) \hat{\omega}_{\text{r}} - \frac{3}{2} p \psi_{\text{f}i_q} \right] \\ \hat{d}_{\text{T}} = \bar{z}_{\text{T}} + l_{\text{T}} \hat{\omega}_{\text{r}} \end{cases} \quad (32)$$

where l_T is the load torque observer gain.

5.2 Stability analysis

From (7), the dynamic error of the load torque observer is expressed as

$$\dot{\bar{e}}_d - \frac{l_T}{I} \bar{e}_d = 0 \quad (33)$$

where

$$\bar{e}_d = \hat{d}_T - \bar{d}_T \quad (34)$$

For stability analysis, the Lyapunov function is defined as

$$V = \frac{1}{2} \bar{e}_d^2 \quad (35)$$

The derivative of the Lyapunov function is obtained in terms of (33) as follows

$$\dot{V} = \bar{e}_d \dot{\bar{e}}_d = \frac{l_T}{I} \bar{e}_d^2(t) \quad (36)$$

5.3 Load torque compensation

$$\dot{\mathbf{x}}_{\text{T}} = A_{\text{T}}\mathbf{x}_{\text{T}} + B_{\text{T}_0}\left(u_{\text{T}} + \frac{B_{\text{T}_1}}{B_{\text{T}_0}}\bar{d}_{\text{T}}\right) \quad (37)$$
$$i_q^* = K_P(\omega_r^* - \hat{\omega}_r) + K_I \int (\omega_r^* - \hat{\omega}_r) dt + \frac{2\bar{d}_T}{3p\psi_s} \quad (38)$$

6 Simulation result

Fig. 4 Block diagram of the proposed LDOBC for sensorless PMSM drive

The back EMF observer gain was determined based on its effects on rotor position and speed estimation accuracy, following which the load torque observer gain was selected according to its load torque estimation response as well as the corresponding error due to the speed estimation error. Moreover, constant V/F control was applied to replace the proposed sensorless vector control below 100 r/min, where the proposed observer was inapplicable due to inverter non-linearities and parameter variation.

6.1 Determination of back EMF observer gain

With proper position error compensation, the estimated speeds under varying back EMF observer gains are shown in Fig. 5b. The fluctuations in estimated speed during the steady-state intensified as the gain l_{e_base} diminished. The simulation results under varying back EMF observer gains confirmed the theoretical analysis of the previous section. To guarantee overall satisfactory control performance of the drive system, the gain l_e should be carefully selected to restrict the compensation angle and the speed fluctuation. In the simulation, $l_{e_base} = -5$.

6.2 Determination of torque observer gain

To determine the adjustment law of torque observer gain, the proposed sensorless LDOBC was conducted with varying torque observer gain. Normally, the smaller the value of l_T , the more accurate and faster the torque estimation. However, the influence of torque observer gain on HF estimation error under sensorless control should also be taken into account. The load torque estimation results considering the speed estimation error under varying torque observer gains are shown in Fig. 5c. The dynamic response of the torque observer declined as torque observer gain increased. The load torque estimation noises, primarily caused by the HF component of the speed estimation error, were restrained as l_T increased. The results of load torque estimation results were confirmed with the speed performance of the LDOBC, shown in Fig. 5d. Therefore, the determination of l_T should balance the maintenance of fast load torque estimation response and the suppression of estimation noise. In this case, $l_T = -1$.

Table 1 Motor specifications

Denotation	Value
stator resistance	0.88 Ω
stator inductance	5.01 mH
number of pole pairs	4
rated speed	2500 r/min
rated current	4.6 A
DC bus voltage	300 V
moment of inertia	0.00106
viscous friction coefficient	0.0001
flux linkage	0.143 Wb

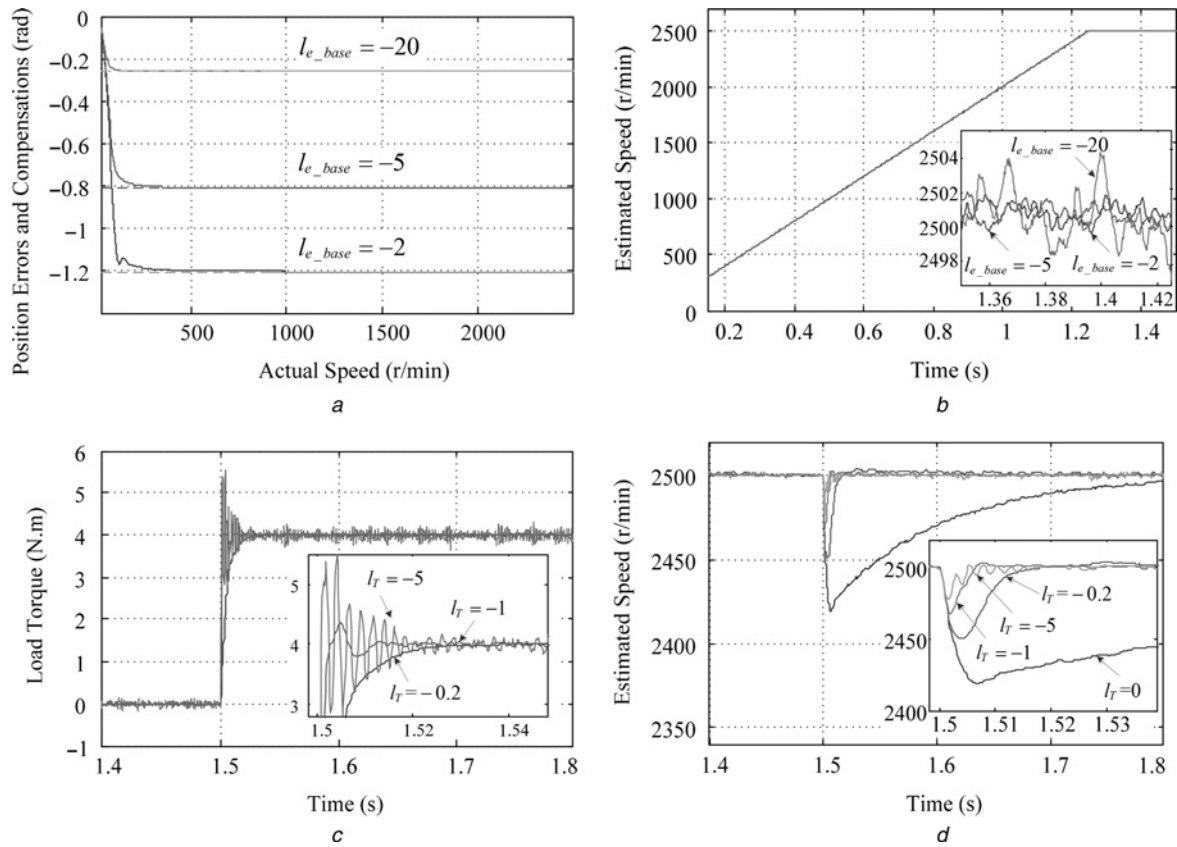


Fig. 5 Simulation results

a Position errors (solid line) and compensation angles (dotted line) against speed under varying back EMF observer gain
b Estimated speeds under varying back EMF observer gain
c Estimated torque considering the speed estimation error under varying torque observer gain
d Estimated speed considering speed estimation error under varying torque observer gain

The determination procedure of observer gain is shown in Fig. 6. The performance of load torque estimation depends on the accuracy of the estimated rotor speed; hence, l_e should be set before l_T . The selection of l_e influences the compensation angle and the speed fluctuation in different directions, whereas the adjustment in l_T needs to balance dynamic responding speed and steady-state torque estimation error.

7 Experimental results

7.1 Experimental setup

The experimental system configuration is shown in Fig. 7. A digital signal processor TMS320F2812 was used for the proposed LDOBC

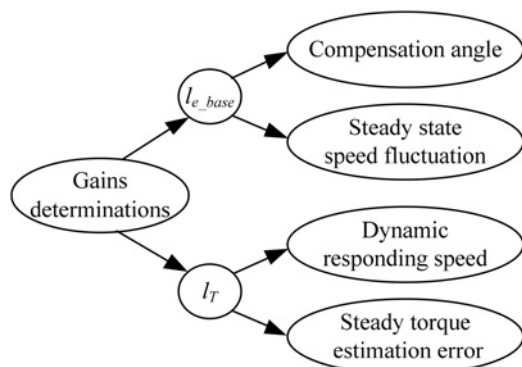


Fig. 6 Gain selection criteria of the observers

for sensorless PMSM drive on a control board. The real-time control software was written in C. The sample time of analogue inputs, which includes stator currents and voltages, was set to 100 μ s. The carrier frequency of the pulse width modulated inverter was 10 kHz.

The parameters of the drive system are listed in Table 1. The inverter on the drive board consisted of six N-Channel Mega IRFP470 metal oxide semiconductor field effect transistors. The tested PMSM, whose rotor position and speed were measured by an encoder for comparison, was connected to a generator whose load torque was adjusted by a tunable load. The base value of the back EMF observer gain was $l_{e_base} = -5$ according to simulation results. Under this condition, the minimum working speed was 100 r/min.

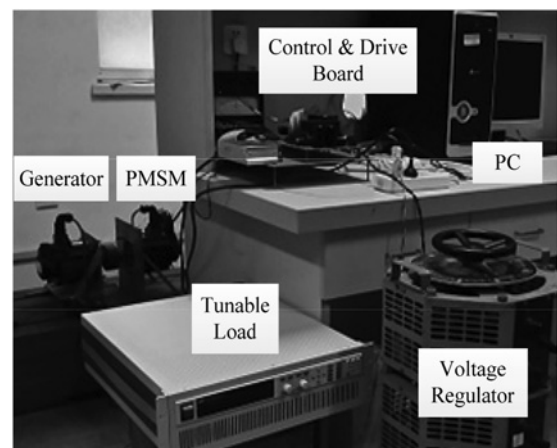


Fig. 7 Experimental setup

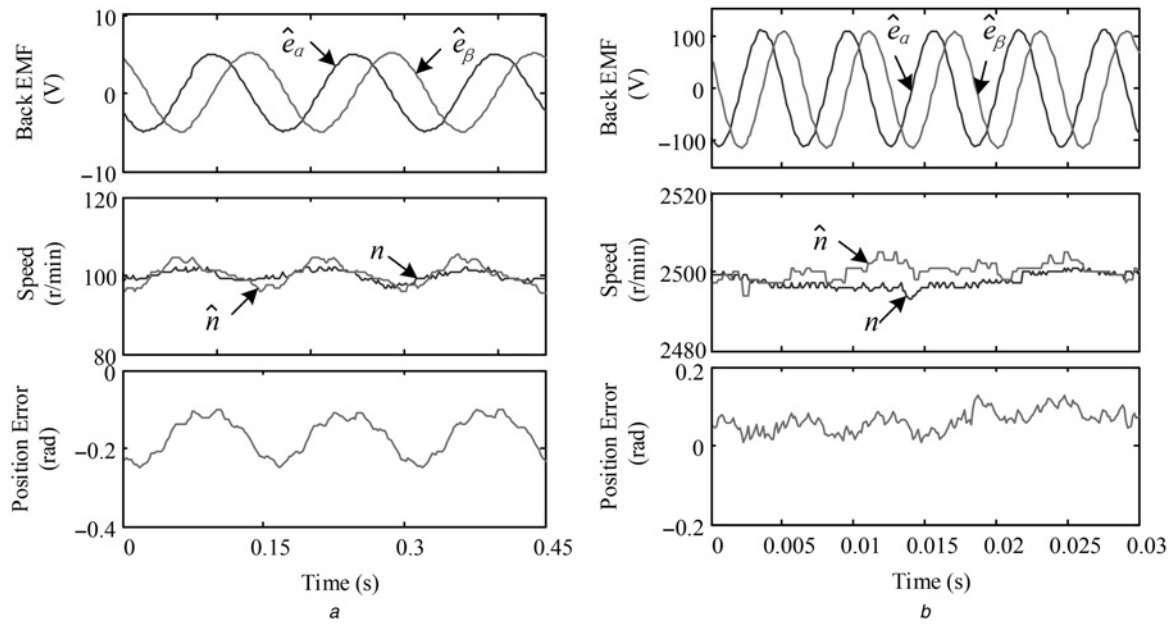


Fig. 8 Experimental results of steady-state sensorless performance

a $n^* = 100$ r/min, DO
b $n^* = 2500$ r/min, DO

7.2 Steady-state sensorless performance

The experimental results of the proposed sensorless method in steady state are shown in Fig. 8. The machine operated under light load conditions with $l_T = 0$. The estimated back EMFs at both 100 and 2500 r/min were close in shape to sinusoidal waveforms and the estimated speed complied with the actual values very well. The effectiveness of the proposed sensorless method in steady state was thus verified.

7.3 Acceleration performance

The dynamic responses of the proposed LDOBC sensorless drive during acceleration with $l_T = 0$ and -1 are shown in Fig. 9. The base value of the back EMF observer gain l_{e_base} was set to -5 . In both cases, the PMSM was accelerated from 500 to 2500 r/min

with 100% rated load in <1 s and the estimated speeds tracked the actual speed very closely. The load torque compensation had little influence on system dynamics during acceleration. The speed error in the transient interval was comparatively larger than that in the steady state, but $B\Delta\omega_{e_DC}$ was negligible considering its amplitude. In general, the position error due to the observer was compensated for. However, a minor DC offset persisted in the position estimation error due to parameter uncertainties at different speeds.

7.4 Load torque rejection due to LDOBC

The dynamics of the sudden torque load step of the proposed sensorless PMSM system with full load are shown in Fig. 10. The descent and ascent of the estimated and the actual speeds at 250 and 800 r/min with $l_T = 0$ during load variation reached 100 and 80 r/min, respectively, and lasted for more than 2 s. On the

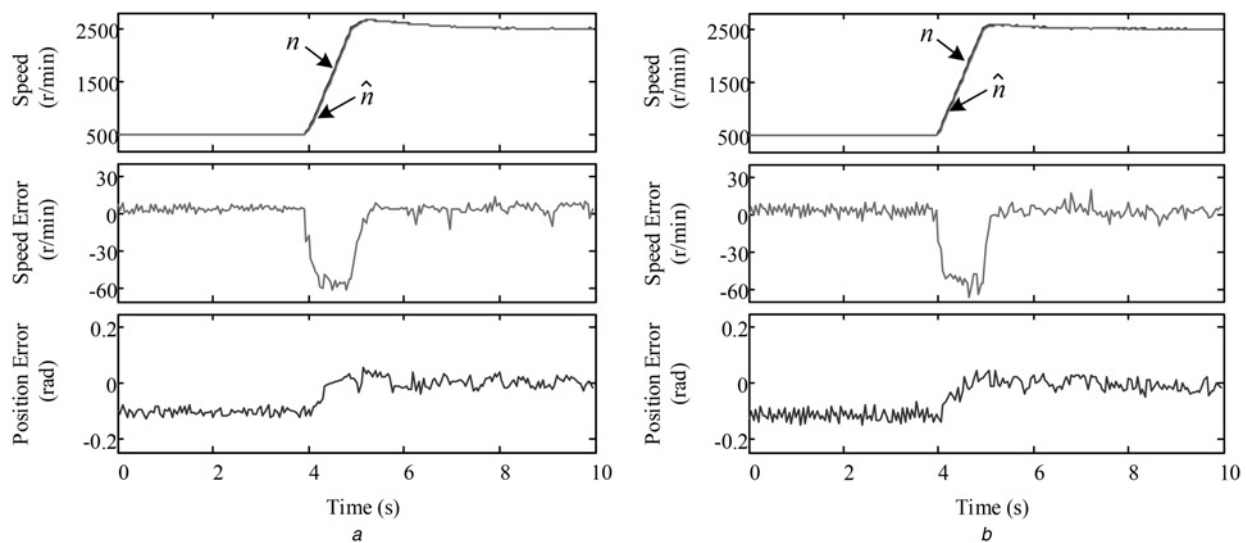


Fig. 9 Experimental results of acceleration from 500 to 2500 r/min with 100% load

a $l_T = 0$
b $l_T = -1$

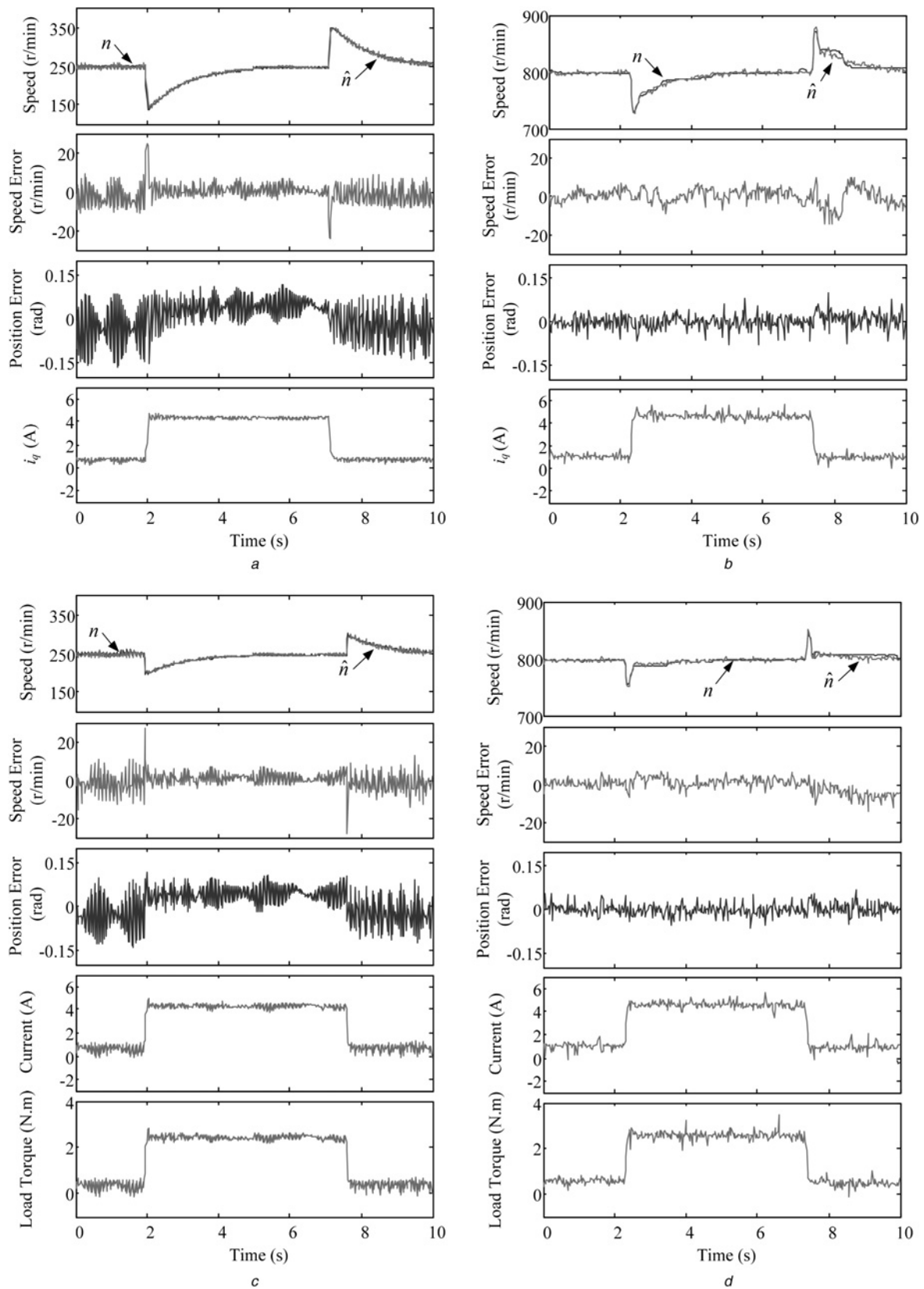


Fig. 10 Experimental results of rated load torque variation

a $I_T = 0$ at 250 r/min
b $I_T = 0$ at 800 r/min
c $I_T = -1$ at 250 r/min
d $I_T = -1$ at 800 r/min

contrary, the perturbations of both the estimated and the actual speeds with $I_T = -1$ were < 50 r/min, and quickly returned to being negligible. Except for the fluctuation of the load torque estimation

at low speed with light load, speed error barely registered an impact on load torque compensation. The selection of the back EMF and load torque observer gains thus proved to be effective

Table 2 Position errors caused by parameter variations

Speed, r/ min	0.5R, L, rad	R, 0.5L, rad	R, L, rad	1.5R, L, rad	R, 1.5L, rad
150	-0.168	0.309	-0.168	-0.161	-0.433
1500	-0.034	-0.296	-0.058	-0.032	-0.237

for the load torque rejection of the proposed sensorless PMSM system.

7.5 Parameter variations

The changes of plant parameters with the operating conditions is inevitable in practical applications. To analyse the impact on the position error of the parameter variations, the average position estimation error under parameter variations at 150 and 1500 r/min are listed in Table 2. The position errors shifted with the inductance variations were more severe than the resistance variations, therefore the proposed sensorless method is more suitable for the resistance-varying applications and its tolerance for the inductance variation is comparatively lower. However, considering general knowledge in the area of the model-based sensorless control, the effects of the stator resistance variation at low speeds are not negligible. The control object of the paper is the surface mounted PMSM, and thus the effect of the parameter error is not severe compared with the case of the IPMSM.

8 Conclusion

This paper proposed a LDOBC for sensorless PMSM drive that is robust against load torque variations. The back EMF-based rotor speed estimator and the load torque observer were both presented according to the design of the disturbance observer. For satisfactory sensorless control, the back EMF observer gain I_e should be carefully selected to simultaneously restrict compensation angle and speed fluctuation. Considering the impacts of the speed estimation error on torque disturbance rejection, the load torque observer gain I_T should maintain fast load torque estimation response and reduce estimation noise due to HF speed estimation error. The adjustment laws for the back EMF and torque observer gains were verified by simulations. Experimental results showed that the proposed rotor speed estimator is accurate for both steady state and dynamic operations under loaded and unloaded conditions. The LDOBC reduced the transient interval of the speed step due to sudden load torque variation, which is suitable for high-performance PMSM applications.

9 References

- Lorenz, R.D.: 'Sensorless as an integration focus for motor drives and power devices'. Proc. Int. Conf. on Electrical Machines and Systems, Seoul, Korea, October 2007, pp. 386–391
- Liu, J., Zhu, Z.: 'Novel sensorless control strategy by high-frequency pulsating signal injection into stationary reference frame', *IEEE Trans. Ind. Appl.*, 2013, **50**, (4), pp. 2574–2583
- Yang, S.-C., Suzuki, T., Lorenz, R.D., *et al.*: 'Surface-permanent-magnet synchronous machine design for saliency-tracking sensorless position estimation at zero and low speeds', *IEEE Trans. Ind. Appl.*, 2011, **47**, (5), pp. 2103–2116
- Park, N.-C., Kim, S.-H.: 'Simple sensorless algorithm for interior permanent magnet synchronous motors based on high-frequency voltage injection method', *IET Electr. Power Appl.*, 2014, **8**, (2), pp. 68–75
- Chen, J.-L., Tseng, S.-K., Liu, T.-H.: 'Implementation of high-performance sensorless interior permanent-magnet synchronous motor control systems using a

- high-frequency injection technique', *IET Electr. Power Appl.*, 2012, **6**, (8), pp. 533–544
- Alonge, F., D'Ippolito, F., Sferlazza, A.: 'Sensorless control of induction-motor drive based on robust Kalman filter and adaptive speed estimation', *IEEE Trans. Ind. Electron.*, 2014, **61**, (3), pp. 1444–1453
- Lee, J., Hong, J., Nam, K., *et al.*: 'Sensorless control of surface-mount permanent-magnet synchronous motors based on a nonlinear observer', *IEEE Trans. Power Electron.*, 2010, **25**, (2), pp. 290–297
- Chan, T.F., Wang, W., Borsje, P., *et al.*: 'Sensorless permanent-magnet synchronous motor drive using a reduced-order rotor flux observer', *IET Electr. Power Appl.*, 2008, **2**, (2), pp. 88–98
- Nguyen, T.D., Foo, G.: 'Sensorless control of a dual-airgap axial flux permanent magnet machine for flywheel energy storage system', *IET Electr. Power Appl.*, 2013, **7**, (2), pp. 140–149
- Nahid-Mobarakeh, B., Meibody-Tabar, F., Sargos, F.-M.: 'Back EMF estimation-based sensorless control of PMSM: robustness with respect to measurement errors and inverter irregularities', *IEEE Trans. Ind. Appl.*, 2007, **43**, (2), pp. 485–494
- Vicente, I., Endemaño, A., Garin, X., *et al.*: 'Comparative study of stabilising methods for adaptive speed sensorless full-order observers with stator resistance estimation', *IET Control Theory Appl.*, 2010, **4**, (6), pp. 993–1004
- Wang, G., Li, Z., Zhang, G., *et al.*: 'Quadrature PLL-based high-order sliding-mode observer for IPMSM sensorless control with online MTPA control strategy', *IEEE Trans. Energy Convers.*, 2013, **28**, (1), pp. 214–224
- Foo, G., Rahman, M.F.: 'Sensorless sliding-mode MTPA control of an IPM synchronous motor drive using a sliding-mode observer and HF signal injection', *IEEE Trans. Ind. Electron.*, 2010, **57**, (4), pp. 1270–1278
- Sayef, S., Foo, G., Rahman, M.F.: 'Rotor position and speed estimation of a variable structure direct-torque-controlled IPM synchronous motor drive at very low speeds including standstill', *IEEE Trans. Ind. Electron.*, 2010, **57**, (11), pp. 3715–3723
- Yan, J., Lin, H., Feng, Y., *et al.*: 'Improved sliding mode model reference adaptive system speed observer for fuzzy control of direct-drive permanent magnet synchronous generator wind power generation system', *IET Renew. Power Gener.*, 2013, **7**, (1), pp. 28–35
- Chen, W.H.: 'Disturbance observer based control for nonlinear systems', *IEEE/ASME Trans. Mechatronics*, 2004, **9**, (4), pp. 706–710
- Yang, J., Chen, W.H., Li, S.: 'Non-linear disturbance observer-based robust control for systems with mismatched disturbances/uncertainties', *IET Control Theory Appl.*, 2011, **5**, (18), pp. 2053–2062
- Li, S.H., Yang, J., Chen, W.H., *et al.*: 'Generalized extended state observer based control for systems with mismatched uncertainties', *IEEE Trans. Ind. Electron.*, 2012, **9**, (12), pp. 4792–4802
- Ohishi, K., Nakao, M., Ohnishi, K., *et al.*: 'Microprocessor-controlled DC motor for load-insensitive position servo system', *IEEE Trans. Ind. Electron.*, 1987, **4**, (1), pp. 44–49
- Yun, J.N., Su, J.B.: 'Design of a disturbance observer for a two-link manipulator with flexible joints', *IEEE Trans. Control Syst. Technol.*, 2014, **22**, (2), pp. 809–815
- Wu, H.N., Liu, Z.Y., Guo, L.: 'Robust L_{∞} -gain fuzzy disturbance observer-based control design with adaptive bounding for a hypersonic vehicle', *IEEE Trans. Fuzzy Syst.*, 2014, **22**, (6), pp. 1401–1412
- Lee, K.-W., Ha, J.-I.: 'Evaluation of back-EMF estimators for sensorless control of permanent magnet synchronous motors', *J. Power Electron.*, 2012, **12**, (4), pp. 605–614
- Kim, J.-S., Sul, S.-K.: 'High performance PMSM drives without rotational position sensors using reduced order observer'. Thirtieth IAS Annual Meeting, IAS '95, Conf. Record of the 1995 IEEE Industry Applications Conf., 1995, 1995, vol. 1, pp. 75–82
- Tomita, M., Senjyu, T., Doki, S., *et al.*: 'New sensorless control for brushless DC motors using disturbance observers and adaptive velocity estimations', *IEEE Trans. Ind. Electron.*, 1998, **45**, (2), pp. 274–282
- Morimoto, S., Kawamoto Sanada, K.M., Takeda, Y.: 'Sensorless control strategy for salient-pole PMSM based on extended EMF in rotating reference frame', *IEEE Trans. Ind. Appl.*, 2002, **38**, (4), pp. 1054–1061
- Chen, Z., Tomita, M., Doki, S., *et al.*: 'An extended electromotive force model for sensorless control of interior permanent-magnet synchronous motors', *IEEE Trans. Ind. Electron.*, 2003, **50**, (2), pp. 288–295
- Solsona, J., Valla, M.I., Muravchik, C.: 'Nonlinear control of a permanent magnet synchronous motor with disturbance torque estimation', *IEEE Trans. Energy Convers.*, 2000, **15**, (2), pp. 163–168
- Sira-Ramirez, H., Linares-Flores, J., Garcia-Rodriguez, C., *et al.*: 'On the control of the permanent magnet synchronous motor: an active disturbance rejection control approach', *IEEE Trans. Control Syst. Technol.*, 2014, **22**, (5), pp. 2056–2063
- Lee, K., Choy, I., Back, J., *et al.*: 'Disturbance observer based sensorless speed controller for PMSM with improved robustness against load torque variation'. 2011 IEEE Eighth Int. Conf. on Power Electronics and ECCE Asia (ICPE ECCE), Jeju, Korea, May 2011, pp. 2537–2543